

DigitMak Portal V1 Technical Specification

Date: 2026-06-15 Status: Draft for architecture review

1. Purpose

DigitMak Portal V1 will be the full one-stop-shop portal for DigitMak at portal.digitmak.mk. It will provide public information, multilingual service access, contact intake, paid offline subscription management, AI first-line help-desk ticketing, real-time ticket chat, internal meeting scheduling, reporting foundations, and compliance/audit controls.

The portal must support the grant obligations for:

- A DigitMak one-stop-shop portal where clients can browse services, schedule meetings, and receive online help-desk services.
- A first-line AI help desk for companies and public institutions.
- Online access through the DigitMak portal and offline face-to-face consultations.
- Evidence and KPI reporting, including AI help-desk subscriptions.
- English, Macedonian, and Albanian access.

2. Locked Product Decisions

- Scope: full DigitMak portal, not only help desk.
- Domain: portal.digitmak.mk.
- Existing site: digitmak.mk remains active; this portal is a subdomain.
- Frontend: React.
- Backend: .NET 10 Web API.
- Database: PostgreSQL.
- Hosting: Linux VM.
- Deployment: GitLab CI/CD, VM details to be specified later.
- Public visitors can submit contact requests only.
- Tickets, real-time chat, and meeting requests require an active personal subscription.
- Users register with email/password.
- Organizations require admin approval.
- Subscriptions are per person, not per organization.
- An organization may have multiple subscribed persons.
- Subscription duration: 12 months.
- Payment: offline, activated manually by admin.
- Calendar: internal calendar for V1.
- Assignee must confirm meeting requests.
- Chat: real-time in V1, implemented with SignalR.

- Moodle: linked later; leave SSO-ready design hooks now.
- Files: VM disk storage initially.
- Macedonian content uses Cyrillic.
- Brevo SMTP is used for email notifications.

3. High-Level Architecture

Use a modular monolith.

3.1 Backend

Technology:

- .NET 10 Web API.
- Entity Framework Core.
- PostgreSQL provider.
- ASP.NET Identity with email/password.
- JWT bearer token authentication.
- Project-specific profile, organization, and subscription tables around ASP.NET Identity.
- SignalR for real-time ticket chat.
- Background worker for email dispatch and scheduled maintenance.
- Brevo SMTP for email delivery.

Recommended backend modules:

- Identity and Access.
- Organizations.
- Subscriptions.
- Public Contact Requests.
- Help Desk Tickets.
- Ticket Chat.
- Internal Calendar and Meetings.
- Service Catalogue and Content.
- File Storage.
- Notifications.
- Audit Logs.
- Reporting and KPI Exports.
- System Settings.

3.2 Frontend

Technology:

- React.
- TypeScript recommended.
- Vite recommended for SPA delivery.
- React Router.
- i18n library such as [i18next](#).
- Role-aware layouts.
- SignalR client for ticket chat.

Frontend areas:

- Public portal.
- Registration/login.
- Client dashboard.
- Staff dashboard.
- Admin dashboard.

3.3 Database

Use PostgreSQL as the transactional store.

Principles:

- Organization and user ownership must be represented explicitly.
- Server-side authorization must never depend only on frontend filtering.
- Audit data should be append-only.
- File metadata is stored in the database; binary files are stored on VM disk.
- All user-visible mutable entities should have created/updated metadata.

3.4 File Storage

Initial storage path:

`/var/lib/digitmak-portal/uploads/{environment}/{entity}/{year}/{month}/{fileId}`

Database stores:

- Original filename.
- Stored filename/path.
- Content type.
- File size.
- Checksum.
- Uploaded by.
- Linked entity type and id.
- Visibility.

- Created timestamp.

This must be abstracted behind a storage service so S3-compatible storage can replace VM disk later.

4. User Roles And Access Model

4.1 Roles

- Public visitor: unauthenticated user.
- Registered client: authenticated user, pending or approved.
- Subscriber: authenticated user with active personal subscription.
- Help-desk agent: staff user handling tickets.
- Expert/advisor: specialist assigned to tickets.
- Admin: full administrative role.

Subscriber should be implemented primarily as an entitlement derived from an active subscription, not only as a static role.

4.2 Access Rules

A user can create tickets, use chat, and request meetings only if all are true:

- Email is verified.
- User account is active.
- User belongs to an approved organization.
- User has an active personal subscription.

Ticket visibility:

- Ticket creator can see their own ticket.
- Assigned help-desk agent can see assigned ticket.
- Assigned expert/advisor can see assigned ticket.
- Admin can see all tickets.
- Other users in the same organization do not see the ticket by default.

Organization visibility:

- A client sees their own organization profile.
- Organization members do not automatically see each other's tickets.
- Admin sees all organizations.
- Staff can see organization data needed for assigned tickets or admin-granted permissions.

5. Main Workflows

5.1 Public Contact Request

1. Public visitor opens contact form.
2. Visitor submits DMA-style intake.
3. System stores **ContactRequest**.
4. System sends confirmation email to submitter.
5. Admin/agent reviews request.
6. Admin/agent may:
 - respond by email,
 - mark as handled,
 - invite registration,
 - invite subscription,
 - link request to an existing organization,
 - convert request into an internal lead record later.

Public contact requests are not tickets and do not provide chat access.

5.2 Registration And Organization Approval

1. User registers with email/password.
2. User verifies email.
3. User creates or joins an organization profile.
4. Organization status starts as **PendingApproval**.
5. Admin reviews organization.
6. Admin approves or rejects organization.
7. Approved users can receive personal subscription invitations.

Organization statuses:

- **PendingApproval**.
- **Approved**.
- **Rejected**.
- **Suspended**.

5.3 Offline Paid Subscription

1. Admin creates a subscription invitation for a user.
2. System emails invitation through Brevo SMTP.
3. User accepts invitation.
4. Subscription status becomes **PendingPayment**.

5. Offline payment occurs outside the system.
6. Admin verifies payment and activates subscription.
7. Subscription starts and expires after 12 months.
8. System sends activation email.
9. Scheduled job expires subscriptions when `expires_at` is reached.

Subscription statuses:

- `Invited`.
- `PendingPayment`.
- `Active`.
- `Expired`.
- `Cancelled`.

5.4 Ticket Workflow

Ticket statuses:

- `New`.
- `Assigned`.
- `InProgress`.
- `Resolved`.
- `Closed`.

Flow:

1. Active subscriber creates a ticket.
2. Ticket starts as `New`.
3. Admin/agent assigns it to help-desk agent and optionally expert/advisor.
4. Status becomes `Assigned`.
5. Assigned staff starts work; status becomes `InProgress`.
6. Client and staff communicate through real-time ticket chat.
7. Staff may add internal notes not visible to client.
8. Final response includes practical recommendation and referral recommendation if relevant.
9. Ticket is marked `Resolved`.
10. Admin/agent closes it as `Closed`.

Referral is recorded as a recommendation in the ticket result, not as an automated external hand-off in V1.

5.5 Real-Time Ticket Chat

Scope:

- Chat belongs to a ticket.
- Only ticket creator, assigned staff, assigned expert/advisor, and admins can participate.
- Messages are persisted.
- SignalR pushes new messages in real time.
- Email notification is sent when recipient is offline or not currently viewing the ticket.

Message types:

- Client message.
- Staff reply.
- Internal note.
- System event.

Internal notes are visible only to staff/admin roles.

5.6 Meeting Scheduling

Meeting statuses:

- Requested.
- Confirmed.
- Rejected.
- Completed.
- Cancelled.

Flow:

1. Subscriber requests meeting from ticket or dashboard.
2. Request includes preferred time or time window.
3. Request is linked to ticket where applicable.
4. Assignee reviews internal calendar.
5. Assignee confirms, rejects, or proposes another time.
6. Confirmed meeting appears in internal calendar.
7. Email notifications are sent to client and assignee.

Meeting types:

- Online.
- Offline / face-to-face.

Fields:

- Subject.
- Description.
- Organization.
- Requested by.
- Assigned staff.
- Linked ticket.
- Meeting type.
- Location or online meeting link.
- Start time.
- End time.
- Status.
- Notes.

6. Recommended Ticket Categories

Initial ticket categories:

- **AI_READINESS**: readiness, maturity, adoption planning.
- **AI_ACT_COMPLIANCE**: first-line AI Act and compliance guidance.
- **AI_USE_CASE**: practical AI use-case discovery.
- **DATA_GOVERNANCE**: data availability, quality, privacy, governance.
- **AUTOMATION_AND_INTELLIGENCE**: ML, GenAI, automation, analytics.
- **DIGITALIZATION_ROADMAP**: roadmap and transformation planning.
- **TEST_BEFORE_INVEST**: labs, pilots, demonstrators, technical validation.
- **TRAINING_AND_SKILLS**: training needs and resource direction.
- **FUNDING_AND_INVESTMENT**: support to find investment.
- **REFERRAL**: AI Factory, DTA, AI-on-Demand, other EDIH.
- **OTHER**: fallback category.

Ticket priorities:

- **Low**.
- **Normal**.
- **High**.
- **Urgent**.

Default priority: **Normal**.

7. Public Contact Form

The public contact form should be based on Digital Maturity Assessment intake, but must remain shorter than a full DMA.

Fields:

- Organization name.
- Organization type:
 - SME.
 - Mid-cap.
 - Public institution.
 - Partner.
 - Other.
- Sector / priority area.
- Municipality / region.
- Website.
- Contact person full name.
- Email.
- Phone.
- Preferred language.
- Number of employees.
- Current digital maturity self-rating, 1 to 5.
- Main need:
 - AI.
 - Digitalization.
 - Training.
 - Funding.
 - Test-before-invest.
 - Other.
- Business challenge or public-service challenge.
- Current software/tools used.
- Current data sources.
- Whether organization already uses AI.
- AI use-case idea, if any.
- Data privacy/compliance concerns.
- Interest in AI Act guidance.
- Training needs.
- Desired timeline.
- Preferred consultation format:
 - Online.
 - Offline.
 - No preference.
- Consent to be contacted.
- Privacy policy acceptance.

Internal DMA category mapping:

- Digital business strategy.
- Digital readiness.
- Human-centric digitalisation.
- Data management.
- Automation and intelligence.
- Green digitalisation.

8. Data Model

8.1 Core Entities

User

- id.
- email.
- email_verified_at.
- password_hash.
- first_name.
- last_name.
- phone.
- preferred_language.
- status.
- created_at.
- updated_at.
- last_login_at.

Role

- id.
- name.

UserRole

- user_id.
- role_id.

Organization

- id.
- name.
- type.
- sector.

- municipality.
- region.
- website.
- employee_count.
- status.
- approved_by.
- approved_at.
- created_at.
- updated_at.

OrganizationMember

- id.
- organization_id.
- user_id.
- member_status.
- is_primary_contact.
- created_at.

SubscriptionInvitation

- id.
- user_id.
- organization_id.
- token_hash.
- status.
- expires_at.
- created_by.
- created_at.
- accepted_at.

Subscription

- id.
- user_id.
- organization_id.
- invitation_id.
- status.
- starts_at.
- expires_at.
- offline_payment_reference.
- payment_note.
- invited_by.

- activated_by.
- activated_at.
- cancelled_by.
- cancelled_at.
- created_at.
- updated_at.

ContactRequest

- id.
- organization_name.
- organization_type.
- sector.
- municipality.
- region.
- website.
- contact_name.
- email.
- phone.
- preferred_language.
- employee_count.
- digital_maturity_rating.
- main_need.
- challenge_description.
- current_tools.
- current_data_sources.
- uses_ai.
- ai_use_case.
- privacy_concerns.
- interested_in_ai_act_guidance.
- training_needs.
- desired_timeline.
- preferred_consultation_format.
- consent_to_contact.
- privacy_policy_accepted.
- status.
- assigned_to.
- linked_organization_id.
- created_at.
- updated_at.

Ticket

- id.
- ticket_number.
- organization_id.
- created_by_user_id.
- category.
- priority.
- status.
- title.
- description.
- assigned_agent_id.
- assigned_expert_id.
- final_recommendation.
- referral_recommendation.
- resolved_at.
- closed_at.
- created_at.
- updated_at.

TicketMessage

- id.
- ticket_id.
- sender_user_id.
- message_type.
- body.
- created_at.
- edited_at.

TicketAttachment

- id.
- ticket_id.
- message_id.
- file_id.
- uploaded_by.
- created_at.

Meeting

- id.
- organization_id.
- requested_by_user_id.
- assigned_user_id.

- ticket_id.
- subject.
- description.
- meeting_type.
- location.
- online_link.
- requested_time_window.
- starts_at.
- ends_at.
- status.
- confirmed_by.
- confirmed_at.
- created_at.
- updated_at.

ServiceCatalogueItem

- id.
- slug.
- status.
- category.
- created_at.
- updated_at.

ContentPage

- id.
- slug.
- status.
- created_at.
- updated_at.

Translation

- id.
- entity_type.
- entity_id.
- language.
- field_name.
- value.

FileObject

- id.
- original_filename.
- stored_path.
- content_type.
- size_bytes.
- checksum.
- visibility.
- uploaded_by.
- created_at.

Notification

- id.
- recipient_user_id.
- type.
- channel.
- subject.
- body.
- status.
- sent_at.
- created_at.

AuditLog

- id.
- actor_user_id.
- actor_ip.
- action.
- entity_type.
- entity_id.
- old_values_json.
- new_values_json.
- metadata_json.
- created_at.

EvidenceFile

- id.
- related_entity_type.
- related_entity_id.
- file_id.
- kpi_category.
- reporting_period.

- `template_type.`
- `created_by.`
- `created_at.`

9. API Surface

All endpoints require HTTPS in production.

9.1 Public

- `GET /api/public/services`
- `GET /api/public/pages/{slug}`
- `POST /api/public/contact-requests`
- `GET /api/public/settings/locales`

9.2 Auth

- `POST /api/auth/register`
- `POST /api/auth/login`
- `POST /api/auth/logout`
- `POST /api/auth/verify-email`
- `POST /api/auth/forgot-password`
- `POST /api/auth/reset-password`
- `GET /api/auth/me`

9.3 Organizations

- `POST /api/organizations`
- `GET /api/organizations/my`
- `PATCH /api/organizations/my`
- `GET /api/admin/organizations`
- `GET /api/admin/organizations/{id}`
- `POST /api/admin/organizations/{id}/approve`
- `POST /api/admin/organizations/{id}/reject`
- `POST /api/admin/organizations/{id}/suspend`

9.4 Subscriptions

- `GET /api/subscriptions/my`
- `POST /api/admin/subscription-invitations`
- `GET /api/admin/subscription-invitations`
- `POST /api/subscription-invitations/{token}/accept`
- `GET /api/admin/subscriptions`
- `POST /api/admin/subscriptions/{id}/activate`

- POST /api/admin/subscriptions/{id}/cancel
- POST /api/admin/subscriptions/{id}/expire

9.5 Contact Requests

- GET /api/admin/contact-requests
- GET /api/admin/contact-requests/{id}
- PATCH /api/admin/contact-requests/{id}
- POST /api/admin/contact-requests/{id}/assign
- POST /api/admin/contact-requests/{id}/mark-handled
- POST /api/admin/contact-requests/{id}/link-organization

9.6 Tickets

- GET /api/tickets/my
- POST /api/tickets
- GET /api/tickets/{id}
- POST /api/tickets/{id}/attachments
- POST /api/tickets/{id}/meetings
- GET /api/staff/tickets
- POST /api/staff/tickets/{id}/assign
- PATCH /api/staff/tickets/{id}/status
- POST /api/staff/tickets/{id}/resolve
- POST /api/staff/tickets/{id}/close
- PATCH /api/staff/tickets/{id}/recommendation

9.7 Ticket Chat

- GET /api/tickets/{id}/messages
- POST /api/tickets/{id}/messages
- POST /api/staff/tickets/{id}/internal-notes
- SignalR hub: /hubs/tickets

SignalR events:

- JoinTicket.
- LeaveTicket.
- TicketMessageCreated.
- TicketStatusChanged.
- TicketAssigned.

9.8 Meetings

- GET /api/meetings/my

- GET /api/staff/meetings
- POST /api/meetings
- POST /api/staff/meetings/{id}/confirm
- POST /api/staff/meetings/{id}/reject
- POST /api/staff/meetings/{id}/complete
- POST /api/meetings/{id}/cancel

9.9 Admin

- GET /api/admin/users
- GET /api/admin/users/{id}
- PATCH /api/admin/users/{id}
- POST /api/admin/users/{id}/roles
- DELETE /api/admin/users/{id}/roles/{role}
- GET /api/admin/audit-logs
- GET /api/admin/reports/subscriptions
- GET /api/admin/reports/tickets
- GET /api/admin/reports/kpis
- GET /api/admin/reports/export

10. Notifications

Email provider: Brevo SMTP.

Required notifications:

- Email verification.
- Password reset.
- Organization submitted for approval.
- Organization approved/rejected.
- Subscription invitation.
- Subscription activated.
- Subscription expiring soon.
- Subscription expired.
- Contact request confirmation.
- Ticket created.
- Ticket assigned.
- Ticket status changed.
- New ticket chat message.
- Meeting requested.
- Meeting confirmed/rejected/cancelled.

Email sending should be queued in the database and sent by background worker.

11. Audit And Compliance

11.1 Audit Events

Audit these actions:

- Login.
- Logout.
- Failed login.
- User registration.
- Email verification.
- Password reset requested/completed.
- User status changed.
- Role added/removed.
- Organization created/updated.
- Organization approved/rejected/suspended.
- Subscription invitation created/accepted.
- Subscription activated/expired/cancelled.
- Contact request created/assigned/handled.
- Ticket created.
- Ticket assigned.
- Ticket status changed.
- Ticket resolved/closed.
- Ticket message created.
- Internal note created.
- File uploaded/deleted.
- Meeting requested/confirmed/rejected/cancelled/completed.
- Content page changed.
- Service catalogue changed.
- System setting changed.

11.2 Data Protection

The portal must include:

- Privacy policy acceptance.
- Terms acceptance.
- Consent to be contacted on public form.
- AI help desk disclaimer.
- Data retention settings.
- Role-based access control.
- Server-side authorization checks.

- Organization and ticket isolation.
- Encrypted secrets.
- Secure password hashing.
- HTTPS-only production deployment.
- JWT expiry, refresh strategy, and revocation behavior must be explicitly implemented and tested.

Disclaimer requirement:

The AI first-line help desk provides initial guidance and recommendations. It does not provide legal certification, formal legal advice, or regulatory approval.

12. Reporting And KPI Foundation

V1 reports:

- Number of active subscriptions.
- Number of expired/cancelled subscriptions.
- Number of AI help-desk subscriptions for KPI tracking.
- Contact requests by organization type, sector, region, need.
- Tickets by category.
- Tickets by status.
- Tickets by assignee.
- Tickets by organization type.
- AI Act guidance requests.
- Referral recommendations by target type.
- Meetings requested/confirmed/completed.
- Public institutions served.
- Export to CSV in V1.

Evidence files:

- Data model included in V1.
- Template-driven evidence creation deferred.
- Upload evidence manually and attach to ticket, meeting, subscription, contact request, or KPI period.

13. Internationalization

Supported languages:

- English.
- Macedonian, Cyrillic.

- Albanian.

Implementation rules:

- Store user preferred language.
- All public pages must be translatable.
- All emails must support templates per language.
- Admin-entered content must support translated fields.
- System enum labels must be translated in frontend.
- Database stores enum codes, not translated labels.

Recommended language codes:

- en.
- mk.
- sq.

14. Frontend Screens

14.1 Public

- Home.
- Services.
- AI Help Desk.
- Training/LMS placeholder.
- Contact.
- Login.
- Register.
- Privacy Policy.
- Terms.

14.2 Client Dashboard

- Dashboard overview.
- My organization.
- My subscription.
- My tickets.
- New ticket.
- Ticket detail with real-time chat.
- My meetings.
- Profile/settings.

14.3 Staff Dashboard

- Assigned tickets.
- Ticket triage.
- Ticket detail.
- Real-time chat and internal notes.
- Meeting requests.
- Calendar.
- Contact requests.
- Client organization view.

14.4 Admin Dashboard

- Users.
- Organizations and approvals.
- Subscription invitations.
- Subscriptions.
- Contact requests.
- Tickets overview.
- Services/content.
- Reports.
- Audit logs.
- System settings.

15. GitLab Repository Structure

Recommended repository:

digitmak-portal/

backend/

src/

tests/

frontend/

src/

tests/

deploy/

docker-compose.yml

nginx/

systemd/

docs/

spec/

If this workspace becomes the repository root, keep the same structure under the current root.

16. GitLab CI/CD

Pipeline stages:

1. `lint_frontend.`
2. `test_frontend.`
3. `build_frontend.`
4. `test_backend.`
5. `build_backend.`
6. `build_images.`
7. `deploy_staging.`
8. `deploy_production.`

Production deployment should be manual and protected.

GitLab protected variables:

- `POSTGRES_PASSWORD.`
- `JWT_SIGNING_KEY.`
- `BREVO_SMTP_HOST.`
- `BREVO_SMTP_PORT.`
- `BREVO_SMTP_USERNAME.`
- `BREVO_SMTP_PASSWORD.`
- `BREVO_FROM_EMAIL.`
- `BREVO_FROM_NAME.`
- `APP_PUBLIC_URL.`
- `ADMIN_BOOTSTRAP_EMAIL.`
- `ADMIN_BOOTSTRAP_PASSWORD.`
- `UPLOADS_ROOT.`

17. VM Deployment Plan

Recommended runtime:

- Linux VM.
- Docker Compose.
- Nginx reverse proxy.
- Let's Encrypt SSL.
- PostgreSQL container or managed local PostgreSQL.
- Backend API container.
- Frontend static container or Nginx serving build output.
- Optional worker process as backend hosted service or separate container.

Recommended containers:

- `frontend`.
- `api`.
- `postgres`.
- `nginx`.

Optional later:

- `redis` for SignalR backplane if scaling beyond one API instance.
- `backup` sidecar.

Backups:

- Daily PostgreSQL dump.
- Daily uploads directory archive.
- Retention policy to be defined.
- Restore procedure must be documented and tested.

18. Security Baseline

Minimum production controls:

- HTTPS.
- HSTS.
- Secure headers in Nginx.
- Rate limiting on login and public contact form.
- Bearer-token storage must be designed to minimize XSS exposure.
- Strong password policy.

- Email verification required.
- Admin-created role changes only.
- No public file listing.
- MIME/type validation for uploads.
- Upload size limits.
- Virus scanning can be deferred but should be planned.
- Audit logs are not editable through UI.
- Secrets only from environment variables or mounted secret files.

19. V1 Milestone Backlog

Milestone 1: Foundation

- Create repository structure.
- Add .NET 10 API project.
- Add React frontend.
- Add PostgreSQL.
- Add Docker Compose for local development.
- Add basic health checks.
- Add GitLab CI skeleton.

Milestone 2: Identity And Organizations

- Email/password auth.
- Email verification.
- Role model.
- Organization creation.
- Admin organization approval.
- Organization-aware authorization.

Milestone 3: Public Portal And Contact Requests

- Public multilingual pages shell.
- Service catalogue.
- Public DMA-style contact form.
- Contact request admin list/detail.
- Contact request email confirmation.

Milestone 4: Subscriptions

- Subscription invitations.
- Invitation email.
- Accept invitation.
- Pending offline payment flow.

- Admin activation.
- Annual expiry.
- Subscriber entitlement checks.

Milestone 5: Tickets

- Subscriber ticket creation.
- Ticket list/detail.
- Assignment.
- Status changes.
- Final recommendation/referral fields.
- Attachments.

Milestone 6: Real-Time Chat

- SignalR ticket hub.
- Ticket chat messages.
- Internal notes.
- Authorization checks for hub access.
- Email notification for new messages.

Milestone 7: Meetings

- Meeting request flow.
- Internal calendar.
- Assignee confirmation.
- Meeting notifications.
- Ticket-linked meetings.

Milestone 8: Audit And Reporting

- Audit log service.
- Audit UI.
- Subscription report.
- Ticket report.
- Contact request report.
- CSV exports.

Milestone 9: Compliance And Polish

- Privacy/terms/disclaimer pages.
- i18n completion.
- Production Nginx config.
- Backup scripts.
- Deployment docs.

- Initial admin bootstrap.

20. Open Items To Resolve Later

- Final VM operating system, CPU, RAM, disk.
- Staging domain.
- Exact GitLab group/project naming.
- Final Brevo SMTP settings.
- File retention period.
- Backup retention period.
- Exact visual design and branding assets.
- Moodle SSO protocol and timeline.
- Whether to add Microsoft/Google calendar integration in V2.
- Evidence templates and structured evidence workflows.